

Quiz 4 - Mid Term Prep

Date: October 3 & 4, 2024

Name: _____

ERP ID: _____

Total Marks: 15 + 5 bonus points

Question 1

[5 points]

Construct a DFA using $\Sigma = \{a, b\}$ which accepts all strings that have an a at the second position and a b at the third last position.

(Construct a DFA, not define)

Question 2**[5 points]**

Prove that the following language $\{a^{n-2}b^nc^{n+2} : n \geq 2\}$ for $\Sigma = \{a, b, c\}$ is not regular.

Question 3**[5 points]**

Construct a DFA using $\Sigma = \{a, b, c\}$ that accepts any string that does not contain the substring "abc".

Question 4: Bonus**[5 points]**

Construct a DFA of the language $\Sigma = \{0, 1\}$ which accepts all the strings with number of 0's in multiples of 5.
(Construct a DFA, not define)